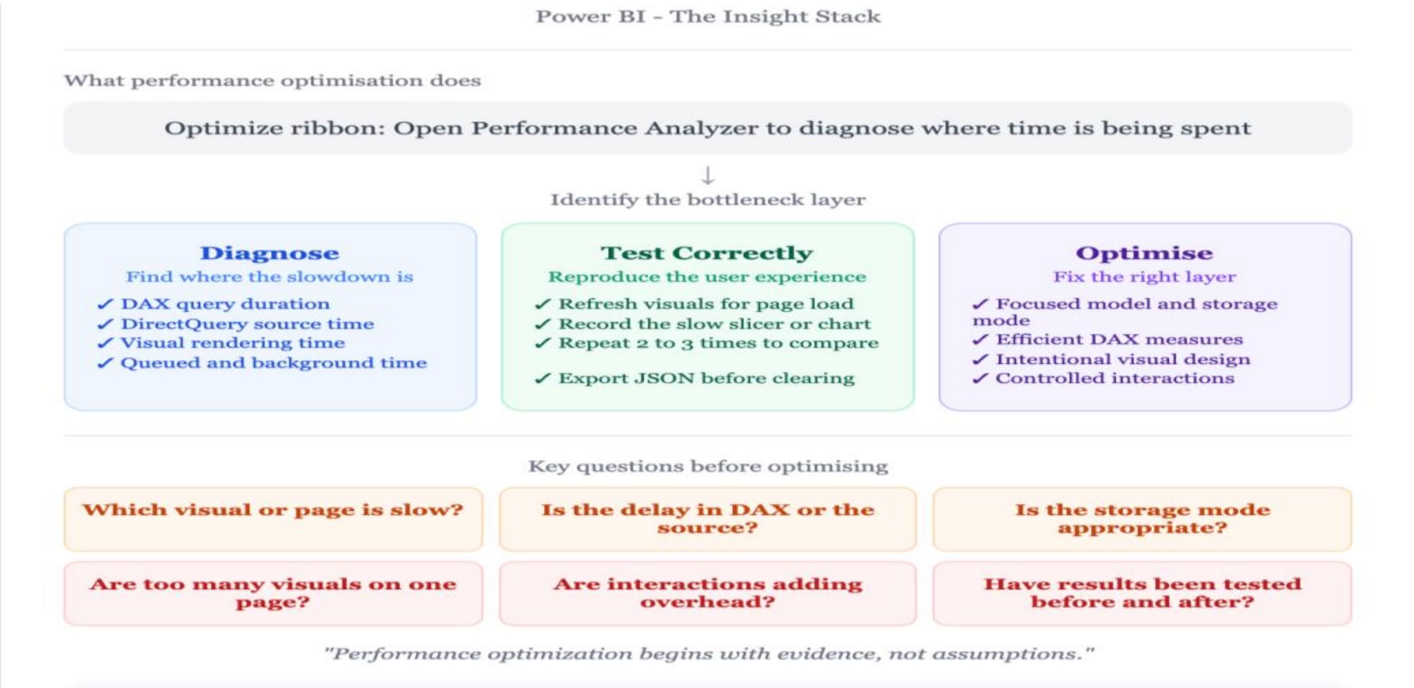
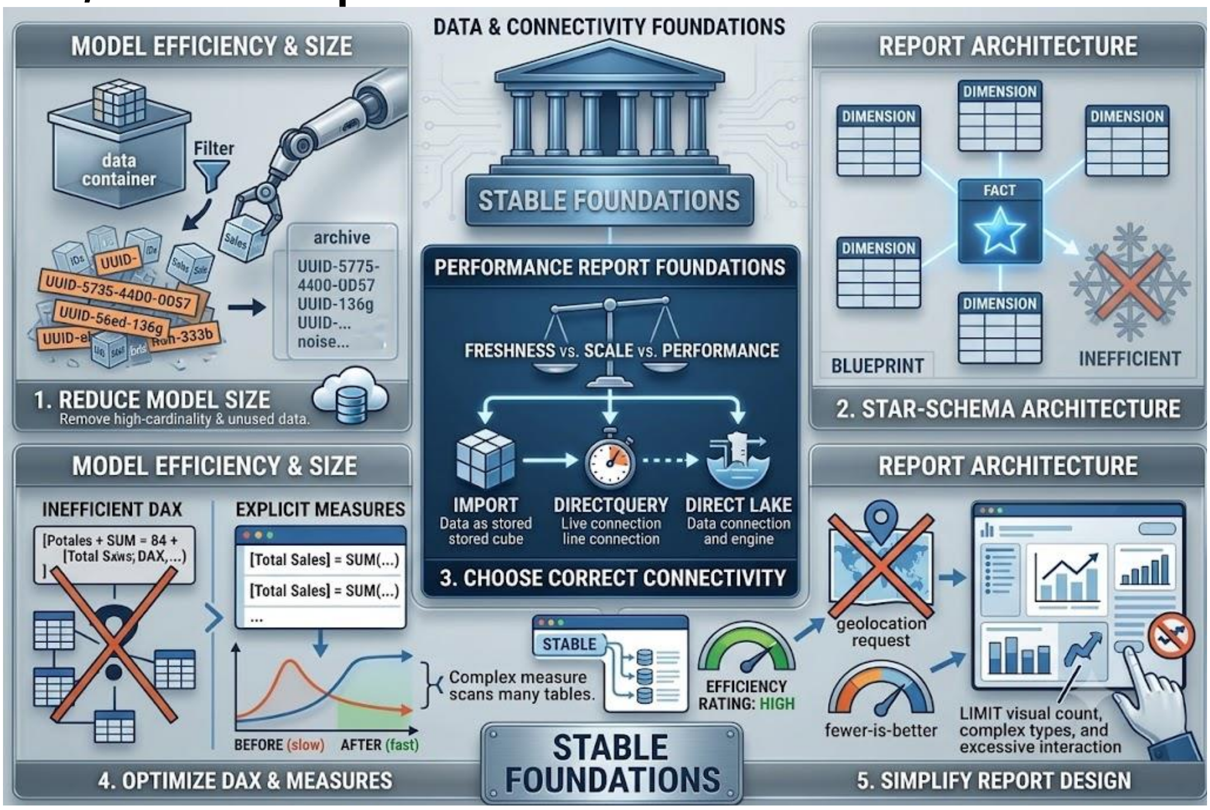


Why a Correct Report Can Still Be Slow

You have connected the right data, cleaned it in Power Query, built a reliable model, written the required DAX, and selected suitable visuals. Yet the report takes too long to open, slicers feel delayed, or a page pauses before it responds. At this point, the question is no longer whether the report is logically correct. The question is where the time is being spent.

Power BI performance is shaped across several layers: the data source, storage mode, semantic model, relationships, DAX, visual design, interactions, gateway, network, and service capacity. Performance Analyzer provides the first report-level diagnostic because it measures how long each visual takes to load and breaks the duration into meaningful components.

First, confirm the performance foundations



Performance Analyzer diagnoses report visuals and interactions. It does not replace model design, Power Query diagnostics, gateway monitoring, or capacity monitoring.

Different Ways to Analyse Power BI Performance

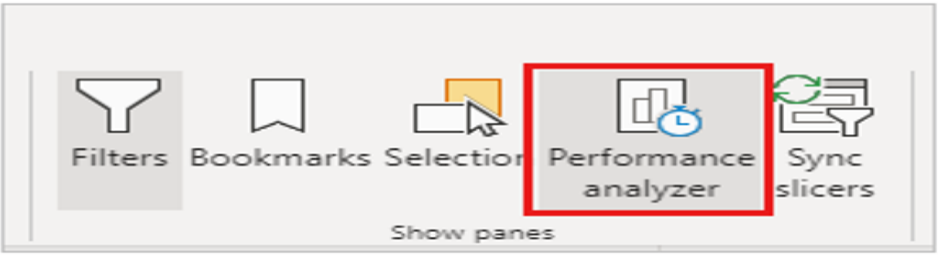
Tool	What it examines	Use it when
Performance Analyzer	Visual load time, DAX query time, DirectQuery time, rendering, and queued/background time.	Start here when a page, visual, slicer, or interaction feels slow.
DAX query view	The DAX query generated for a visual and the result returned by the semantic model.	Use when DAX query duration is high or the result needs validation.
Query Diagnostics	Power Query activity during authoring and refresh.	Use when data preview, transformations, or refresh are slow.
Fabric Capacity Metrics	Capacity utilization, throttling, and workload pressure.	Use when reports are fast locally but slow on Fabric/Premium capacity.
Gateway monitoring	On-premises data transfer, CPU, memory, and gateway bottlenecks.	Use when cloud reports depend on on-premises sources.

Performance Analyzer:

Performance Analyzer is a pane available in Power BI Desktop Report view and while editing a report in the Power BI service. It records how long report visuals take to load or reload after actions such as opening a page, changing a slicer, selecting a data point, or refreshing visuals. For each recorded interaction, the pane lists the visuals involved, their total duration in milliseconds, and the components that contributed to that duration. It can also copy a visual's DAX query, open it in DAX query view in Desktop, refresh one visual, refresh the current page, and export the captured log to JSON.

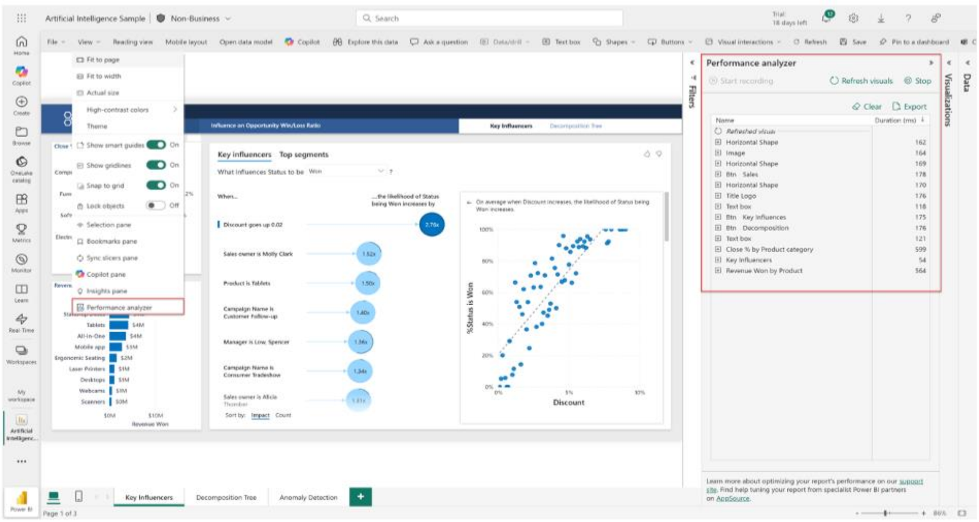
In Power BI Desktop

1. Open the report in Power BI Desktop and go to Report view.
2. Select the Optimize ribbon.
3. Select Performance Analyzer to open the pane on the right side of the canvas.



In the Power BI service

1. Open the report in the Power BI service.
2. Select Edit. You need permission to edit the report.
3. Open the View menu and select Performance Analyzer.



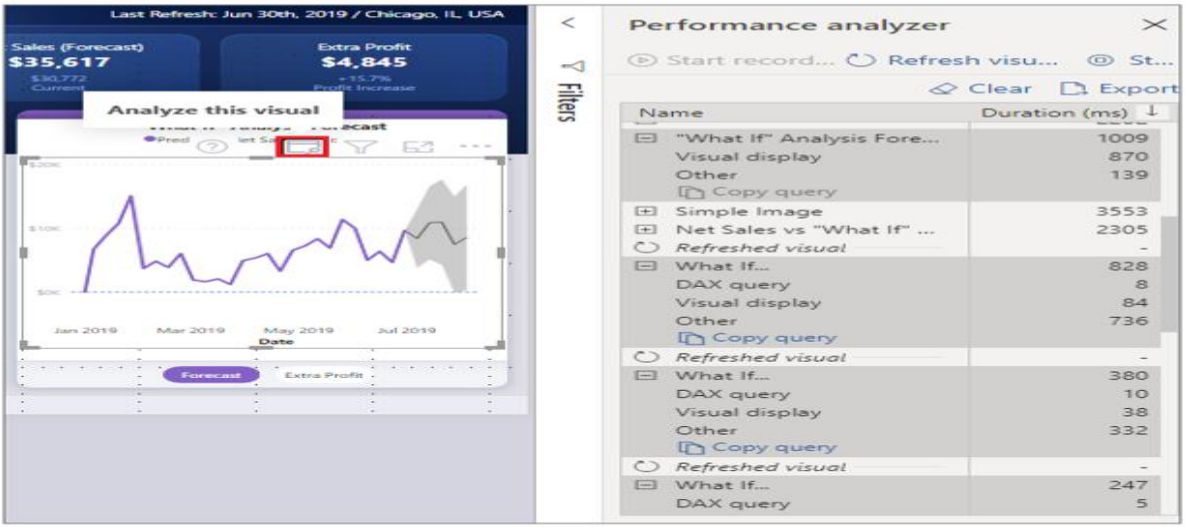
Start a repeatable recording

1. Open the page or report state you want to test.
2. Select Start recording.
3. Select Refresh visuals for a consistent page-level test, or perform the exact interaction users say is slow.
4. Repeat the same action two or three times and compare results. Caching, network conditions, and queued operations can create variation.
5. Select Stop when the test is complete. Do not select Clear until you have reviewed or exported the results.

Capture the Right Performance Scenario

Performance testing should reproduce the user experience. If users complain that the page loads slowly, refresh the page. If a slicer is slow, record that slicer change. If one chart is slow, analyse that visual separately. A broad page refresh and a targeted visual refresh answer different questions.

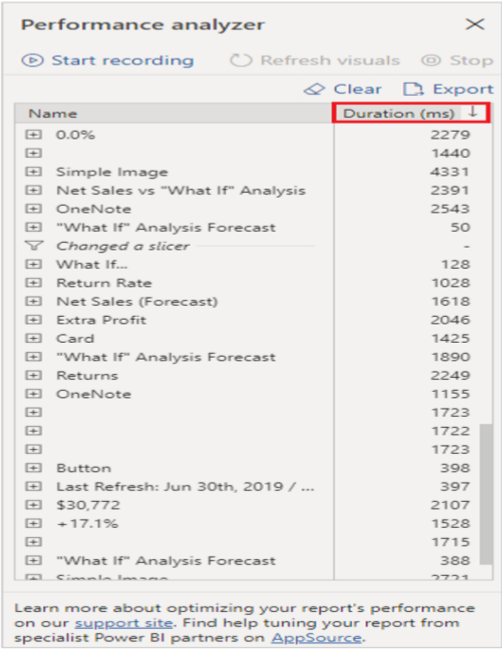
- Refresh visuals: records all visuals on the current page.
- Analyse this visual: refreshes and captures only the selected visual while recording.
- User interaction: records slicer changes, selections, cross-filtering, and navigation-related reloads.



What the Performance Analyzer Results Mean

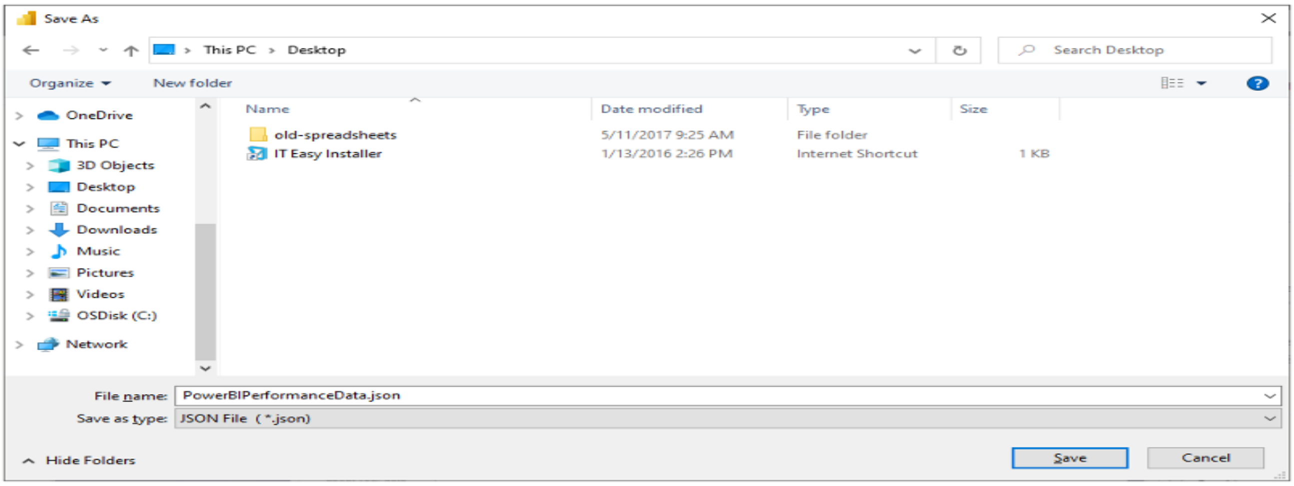
Each visual has a total Duration value and a breakdown of the work involved. The largest value is the first clue, not the final diagnosis. Expand the visual, identify the dominant category, and then investigate the relevant layer.

Signal	What it usually means	First action
DAX query	Time from the visual query until the semantic model returns the result.	Review the measure, filter context, scans, relationships, and generated query.
Direct query	Time for the external DirectQuery request to return.	Review source performance, query folding, gateway/network latency, concurrency, and storage mode.
Visual display	Time required to draw the visual, including web images or geocoding.	Simplify the visual, reduce data points, test a native visual, or reduce map/image work.
Other	Query preparation, waiting for other visuals, or background work. It can include queue time on the shared UI thread.	Reduce page visuals, split content across pages, limit interactions, and test the visual alone.
Evaluated parameters	Time spent evaluating field parameters.	Simplify parameter logic or test a static field selection.



Save the Performance Analyzer Results

Select Export in the Performance Analyzer pane to save the current log as a JSON file. Export before selecting Clear, because Clear permanently removes the captured entries from the pane.

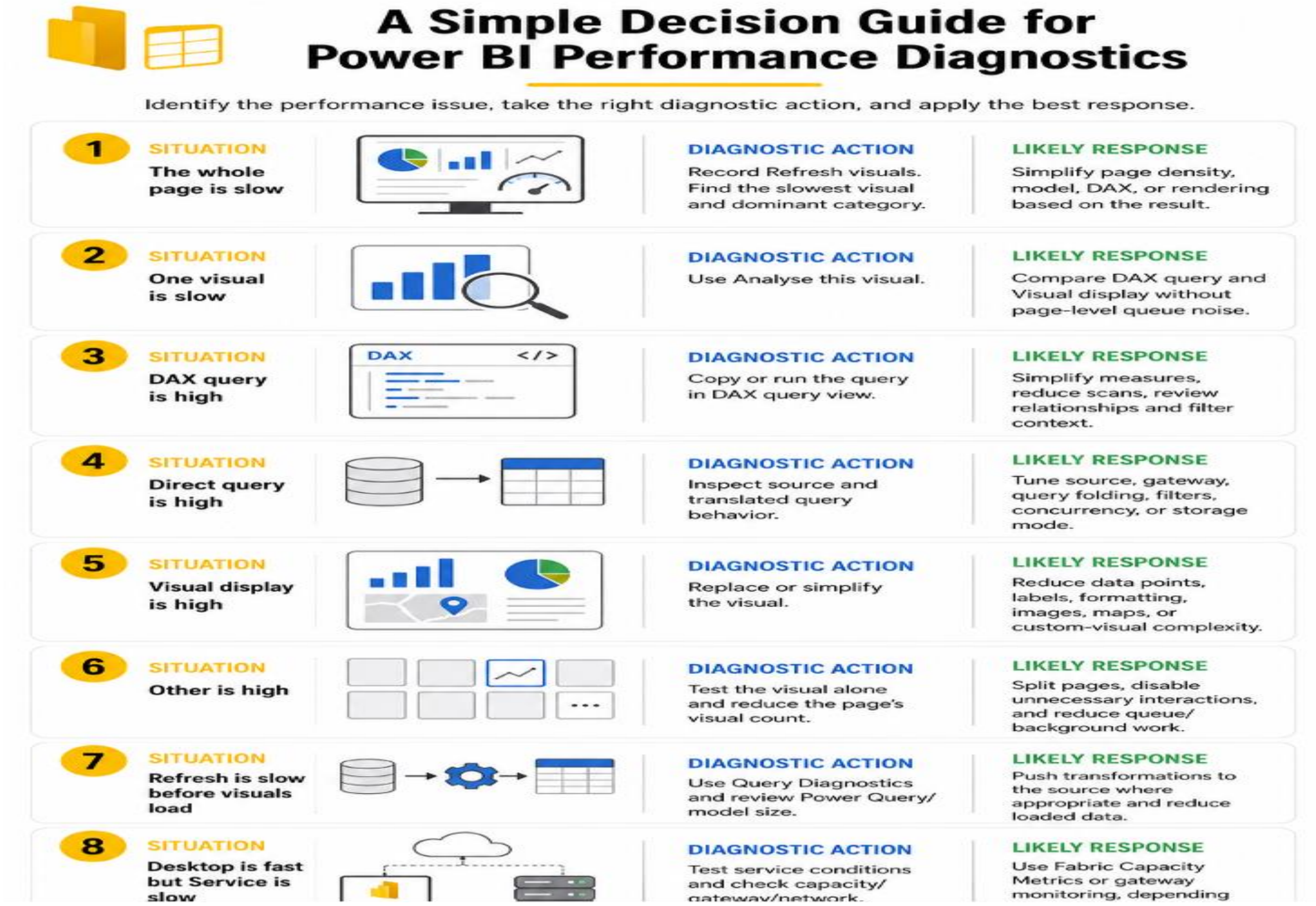


- Use a consistent file name, such as Report_Page_Interaction_Before.json and Report_Page_Interaction_After.json.
- Record the Power BI Desktop version, date, storage mode, selected filters, and test action so results can be compared fairly.
- Keep the JSON with the project's test evidence or optimization notes when performance changes need to be reviewed by another analyst.

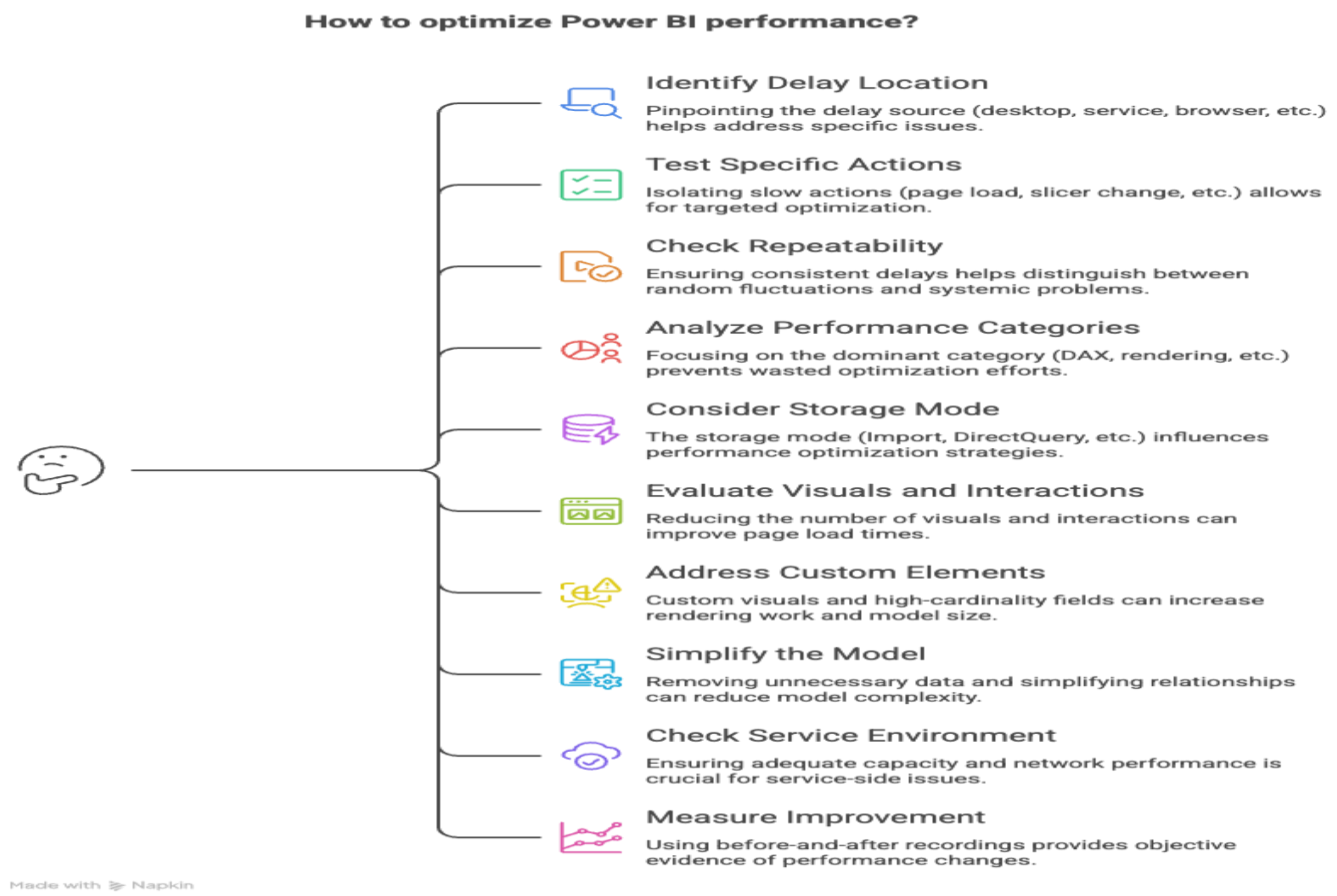
Advantages of Performance Analyzer

- Pinpoints exact bottlenecks instead of relying on assumptions.
- Separates semantic-model query time, DirectQuery source time, visual rendering, and queued/background time.
- Captures real user interactions such as slicer changes and visual selections.
- Provides the DAX query generated by a visual for deeper investigation.
- Supports before-and-after comparison after optimization changes.
- Improves end-user experience by focusing effort on the actual source of delay.
- Exports evidence to JSON for documentation and deeper analysis.

Practical Way to Decide



What Analysts Should Ask



Conclusion

Performance optimization begins with evidence. Performance Analyzer shows whether a slow report is being delayed by DAX, DirectQuery, visual rendering, field parameters, or queued background work. Once the bottleneck is visible, the analyst can make a focused change instead of rebuilding the report blindly.

A fast Power BI report is not created by one setting. It comes from a focused model, appropriate connectivity, efficient DAX, intentional visual design, controlled interactions, and a repeatable test-and-improve process.

Stay tuned for the next article in The Insight Stack: Power BI, Layer by Layer, where we move into **Week 11: Security and Governance** and explore how analysts control access, protect sensitive information, and publish trusted Power BI content

Further Reading:

- Level up your reports: **Mastering Power BI Performance Analyzer | MIcrosoft Fabric Community**
- Guy in a Cube : **Review Performance analyser | Youtube**
- Power BI Report Optimization : **Tabular Editor | Kurt Buhler**